Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	3 July 2026
Team ID	SWUID2026212002
Project Name	Smart Lender
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/

References:

https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web interface for loan application and prediction.	HTML, CSS, JavaScript, React.js
2.	Application Logic-1	User registration, login and authentication.	Python (Flask)
3.	Application Logic-2	Loan application processing and validation.	Python, Pandas
4.	Application Logic-3	AI-based loan approval prediction.	Scikit-learn, NumPy
5.	Database	Store user, application and prediction details.	MySQL
6.	Cloud Database	Cloud-based database service.	Firebase / MySQL Cloud
7.	File Storage	Store uploaded documents and reports.	Local File System / Firebase Storage
8.	External API-1	Email notification service.	SMTP / Email API
9.	External API-2	OTP verification service.	Twilio OTP API
10.	Machine Learning Model	Loan Approval Prediction Model.	Random Forest Classifier (Scikit-learn)
11.	Infrastructure (Server / Cloud)	Application deployment on server / cloud.	Local Server / Render / Railway / AWS

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used.	React.js, Flask, Scikit-learn
2.	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	SHA-256, HTTPS
3.	Scalable Architecture	Justify the scalability of architecture (3 – tier, Micro-services).	Three-Tier Architecture
4.	Availability	Justify the availability of application (e.g. use of load balancers, distributed servers etc.).	Cloud Hosting
5.	Performance	Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN's) etc.	Pandas, NumPy, Optimized ML Model